

WEEK 3

Stock Data

The screenshot shows the Power BI Desktop interface with the 'Table tools' ribbon active. The ribbon includes options for 'Manage relationships', 'New measure', 'Quick measure', 'New column', 'New table', and 'Mark as date table'. Below the ribbon, a data table is displayed with columns: date, open, high, low, close, adj_close, volume, and date - Copy. The table contains 30 rows of stock data for the period from May 21, 2015, to June 17, 2015. The 'Data' pane on the right shows the table structure and available fields for visualization.

date	open	high	low	close	adj_close	volume	date - Copy
21-05-2015 00:00:00	2.79999995231628	2.87400007247925	2.41100001335144	2.56800007820129	2.56800007820129	123039000	21 May 2015
22-05-2015 00:00:00	2.60700011253357	3.10999989509583	2.5999999463257	2.83100008964539	2.83100008964539	28412000	22 May 2015
26-05-2015 00:00:00	2.98000001907349	3.03399991989136	2.90799999237061	2.96499991416931	2.96499991416931	8202000	26 May 2015
27-05-2015 00:00:00	3.06699991226196	3.08100008964539	2.70000004768372	2.75	2.75	7976000	27 May 2015
28-05-2015 00:00:00	2.75500011444092	2.7739999294281	2.64800000190735	2.74499988555908	2.74499988555908	4000000	28 May 2015
29-05-2015 00:00:00	2.72499990463257	2.73799991607666	2.68700003623962	2.72000002861023	2.72000002861023	2554000	29 May 2015
01-06-2015 00:00:00	2.72000002861023	2.74399995803833	2.66000008583069	2.73300004005432	2.73300004005432	2569000	01 June 2015
02-06-2015 00:00:00	2.74699997901917	2.74699997901917	2.66499996185303	2.6949999332428	2.6949999332428	3798000	02 June 2015
03-06-2015 00:00:00	2.67600011825562	2.75	2.65000009536743	2.74399995803833	2.74399995803833	3024000	03 June 2015
04-06-2015 00:00:00	2.74200010296683	2.74900007247925	2.55100011825562	2.67799997329712	2.67799997329712	3226000	04 June 2015
05-06-2015 00:00:00	2.66599988937378	2.73900008201599	2.60999989509583	2.73900008201599	2.73900008201599	1294000	05 June 2015
08-06-2015 00:00:00	2.69000005722046	2.74799990653992	2.68199992179871	2.69799995422363	2.69799995422363	2809000	08 June 2015
09-06-2015 00:00:00	2.66400003433228	2.84599995613098	2.66400003433228	2.83400011062622	2.83400011062622	3375000	09 June 2015
10-06-2015 00:00:00	2.84500002861023	2.8880001144409	2.79500007629395	2.80999994277954	2.80999994277954	2734000	10 June 2015
11-06-2015 00:00:00	2.84899997711182	3.07299995422363	2.84899997711182	3.07299995422363	3.07299995422363	4606000	11 June 2015
12-06-2015 00:00:00	3.13199996948242	3.36999988555908	3.125	3.36299991607666	3.36299991607666	8474000	12 June 2015
15-06-2015 00:00:00	3.3970000743866	3.8120000623962	3.29200005531311	3.54999995231628	3.54999995231628	16943000	15 June 2015
16-06-2015 00:00:00	3.6059994659424	3.8378999232458	3.58400011062622	3.81100010871887	3.81100010871887	11095000	16 June 2015
17-06-2015 00:00:00	3.963000055912781	4.21299982070923	3.80100011825562	3.86299991607666	3.86299991607666	15952000	17 June 2015
18-06-2015 00:00:00	3.8420000762939	3.88899993896484	3.4326000213623	3.5550000667572	3.5550000667572	11482000	18 June 2015
19-06-2015 00:00:00	3.63299989700317	3.67400002479553	3.25300011444092	3.55200004577637	3.55200004577637	9797000	19 June 2015
22-06-2015 00:00:00	3.67600011825562	3.84999990463257	3.60899996757507	3.69400000572205	3.69400000572205	7922000	22 June 2015
23-06-2015 00:00:00	3.77500009536743	3.80999994277954	3.5169999595468	3.54299998283386	3.54299998283386	2939000	23 June 2015
24-06-2015 00:00:00	3.51200008392334	3.74289989471436	3.41400003433228	3.59599995613098	3.59599995613098	5644000	24 June 2015
25-06-2015 00:00:00	3.6340000629425	3.65000009536743	3.48000001907349	3.48399996757507	3.48399996757507	2082000	25 June 2015
26-06-2015 00:00:00	3.4669998855591	3.53200006484985	3.26300001144409	3.3519999809265	3.3519999809265	3465000	26 June 2015
29-06-2015 00:00:00	3.252000092346008	3.3510000705719	3.11999988555908	3.13199996948242	3.13199996948242	3155000	29 June 2015

The Shopify stock dataset was imported into Power BI and reviewed in Data view. The dataset contains important stock-market fields such as Date, Open, High, Low, Close, Adjusted Close, and Volume. These columns provide the required information for analysing stock price movement and trading activity.

Date Table

The screenshot shows the Power BI Desktop interface with the 'Table tools' ribbon active. The ribbon includes options for 'Manage relationships', 'New measure', 'Quick measure', 'New column', 'New table', and 'Mark as date table'. Below the ribbon, a date table is displayed with columns: Date, Year, Quarter, Month, Month Number, and Day. The table contains 30 rows of date data for the period from May 21, 2015, to June 17, 2015. The 'Data' pane on the right shows the table structure and available fields for visualization.

Date	Year	Quarter	Month	Month Number	Day
21-05-2015 00:00:00	2015	Q2	May	5	21
22-05-2015 00:00:00	2015	Q2	May	5	22
23-05-2015 00:00:00	2015	Q2	May	5	23
24-05-2015 00:00:00	2015	Q2	May	5	24
25-05-2015 00:00:00	2015	Q2	May	5	25
26-05-2015 00:00:00	2015	Q2	May	5	26
27-05-2015 00:00:00	2015	Q2	May	5	27
28-05-2015 00:00:00	2015	Q2	May	5	28
29-05-2015 00:00:00	2015	Q2	May	5	29
30-05-2015 00:00:00	2015	Q2	May	5	30
31-05-2015 00:00:00	2015	Q2	May	5	31
01-06-2015 00:00:00	2015	Q2	Jun	6	1
02-06-2015 00:00:00	2015	Q2	Jun	6	2
03-06-2015 00:00:00	2015	Q2	Jun	6	3
04-06-2015 00:00:00	2015	Q2	Jun	6	4
05-06-2015 00:00:00	2015	Q2	Jun	6	5
06-06-2015 00:00:00	2015	Q2	Jun	6	6
07-06-2015 00:00:00	2015	Q2	Jun	6	7
08-06-2015 00:00:00	2015	Q2	Jun	6	8
09-06-2015 00:00:00	2015	Q2	Jun	6	9
10-06-2015 00:00:00	2015	Q2	Jun	6	10
11-06-2015 00:00:00	2015	Q2	Jun	6	11
12-06-2015 00:00:00	2015	Q2	Jun	6	12
13-06-2015 00:00:00	2015	Q2	Jun	6	13
14-06-2015 00:00:00	2015	Q2	Jun	6	14
15-06-2015 00:00:00	2015	Q2	Jun	6	15
16-06-2015 00:00:00	2015	Q2	Jun	6	16
17-06-2015 00:00:00	2015	Q2	Jun	6	17

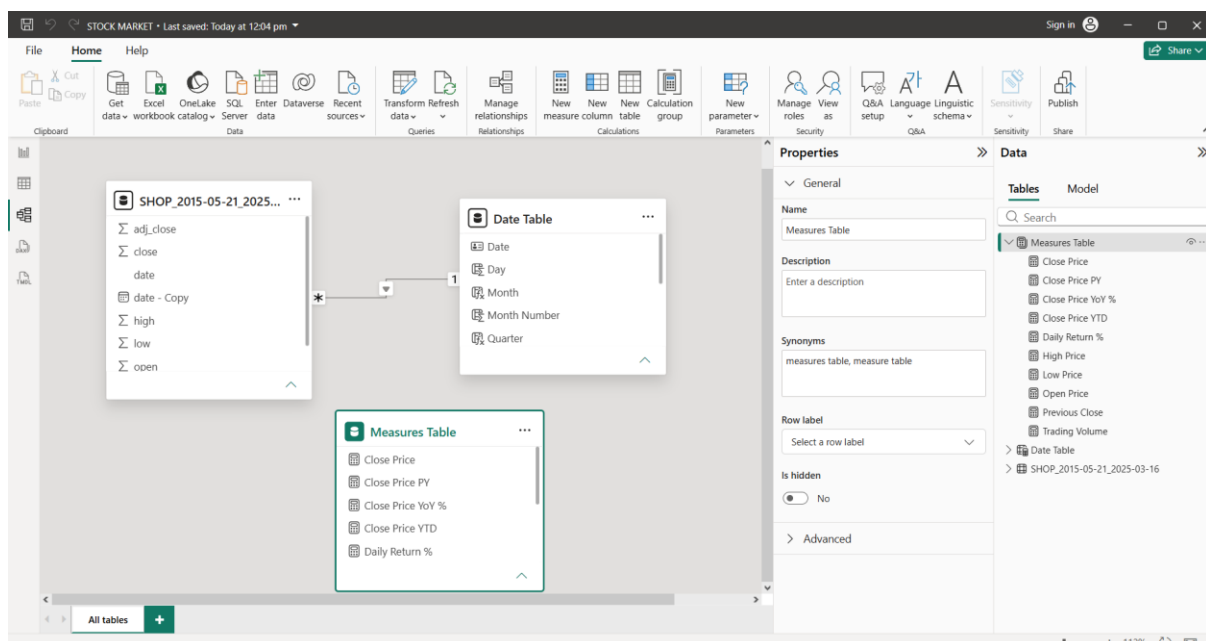
A separate Date Table was created using the CALENDAR() DAX function. The table contains Date, Year, Quarter, Month, Month Number, and Day columns. These fields were used to create a Year → Quarter → Month → Day hierarchy for time-based analysis and drill-down functionality.

DAX used:

Date Table =

```
CALENDAR(
    MIN('SHOP_2015-05-21_2025-03-16'[Date]),
    MAX('SHOP_2015-05-21_2025-03-16'[Date])
)
```

Measures Table



A separate Measures Table was created to organize all the DAX measures used in the dashboard. The table contains financial measures such as Open Price, Close Price, High Price, Low Price, Trading Volume, and Daily Return %. It also includes time-intelligence measures such as Previous Close, Close Price PY, Close Price YoY %, and Close Price YTD. The Measures Table is kept disconnected from the other tables and is used only for storing and organizing measures.

- Close Price = AVERAGE('SHOP_2015-05-21_2025-03-16'[close])

- Close Price PY = CALCULATE([Close Price], SAMEPERIODLASTYEAR('Date Table'[Date]))
- Close Price YoY % = DIVIDE([Close Price] - [Close Price PY], [Close Price PY], 0)
- Close Price YTD =

TOTALYTD(

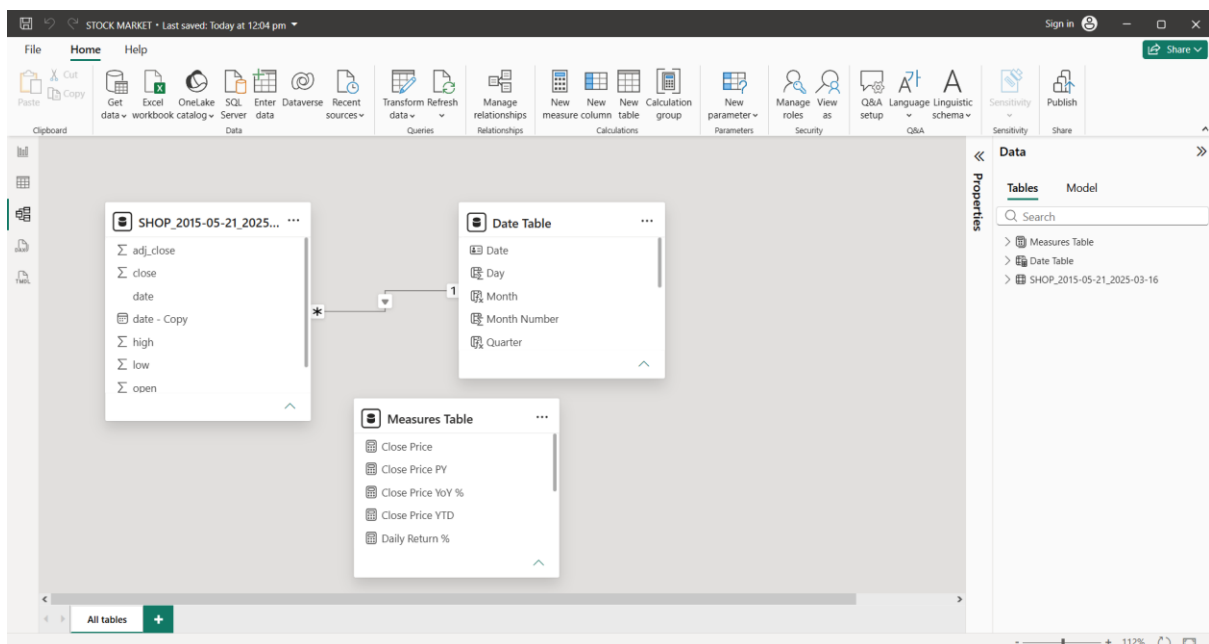
[Close Price],

'Date Table'[Date]

)

- Daily Return % = DIVIDE([Close Price] - [Previous Close], [Previous Close], 0)
- High Price = AVERAGE('SHOP_2015-05-21_2025-03-16'[high])
- Low Price = MIN('SHOP_2015-05-21_2025-03-16'[low])
- Open Price = AVERAGE('SHOP_2015-05-21_2025-03-16'[open])
- Previous Close = CALCULATE([Close Price], DATEADD('Date Table'[Date], -1, DAY))
- Trading Volume = SUM('SHOP_2015-05-21_2025-03-16'[volume])

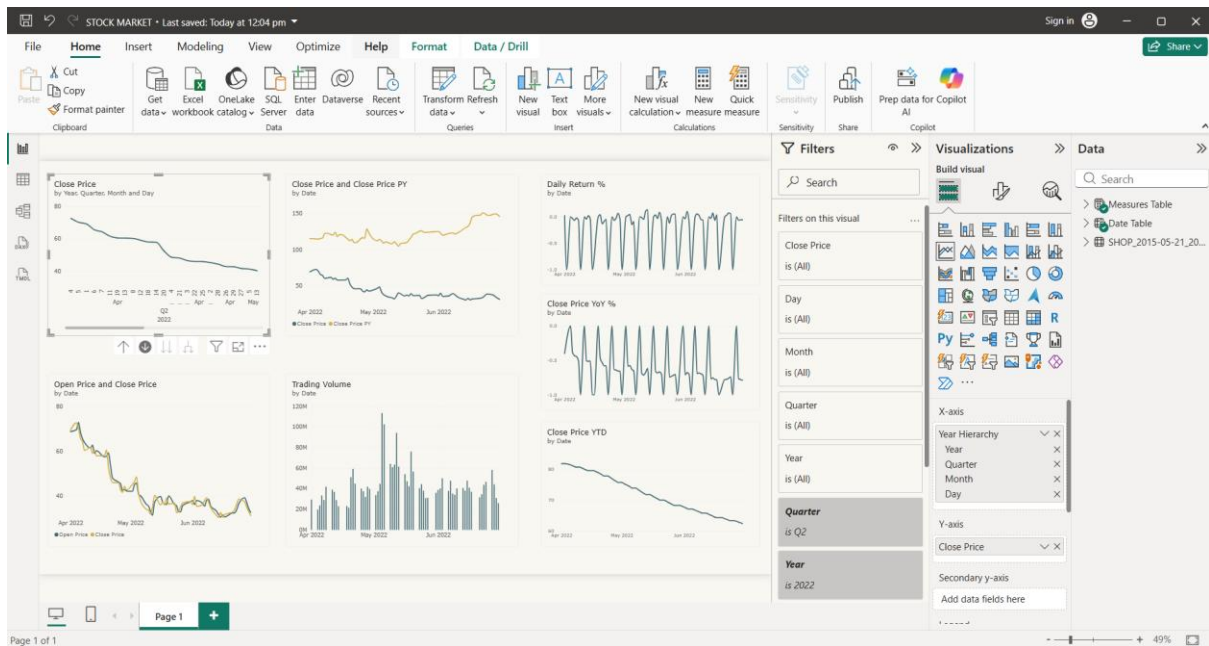
Relationship / Model View



The Model view shows the relationship between the Date Table and the Shopify stock data table. A one-to-many relationship was established between Date

Table[Date] and the stock data date column. The relationship is active and uses single-direction filtering. The Measures Table remains disconnected as it is only used as a container for DAX measures.

Stock Market Dashboard



An interactive stock-market dashboard was created to analyse Shopify stock performance over time. The dashboard includes Close Price, Open Price and Close Price comparison, Close Price versus Previous Year, Daily Return %, Close Price YoY %, Trading Volume, and Close Price YTD. A Year → Quarter → Month → Day hierarchy was also implemented to support drill-down analysis at different levels of time.

Drill-down:

The Close Price chart allows the user to analyse stock prices from **Year** → **Quarter** → **Month** → **Day**, providing detailed time-based analysis of stock-price movement.